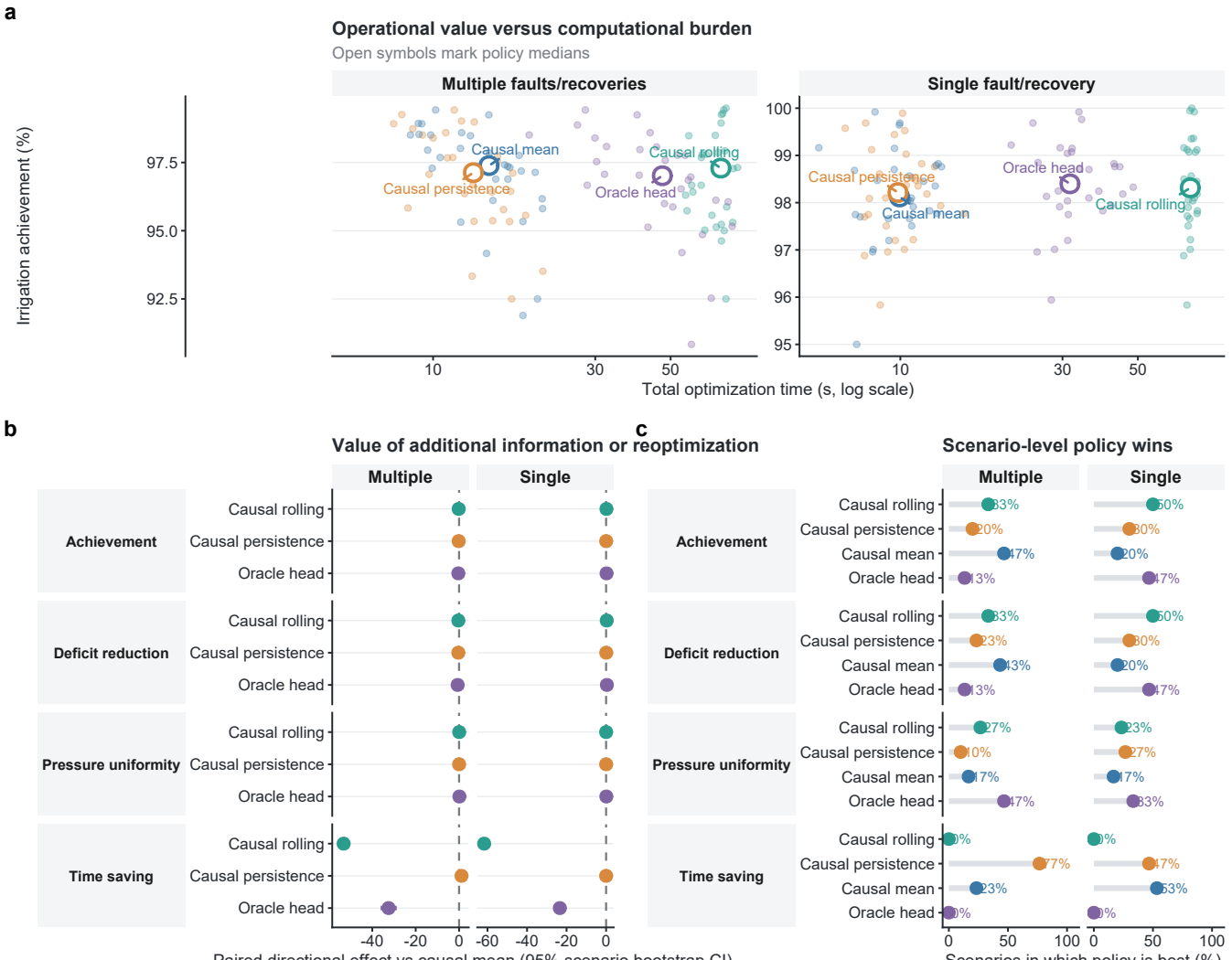
a
Operational value versus computational burden
Open symbols mark policy medians
Multiple faults/recoveries
Single fault/recovery
Irrigation achievement (%)
97.5
95.0
92.5
100
99
98
97
96
95
Causal mean
Causal rolling
Causal persistence
Oracle head
10
30
50
Total optimization time (s, log scale)
b
Value of additional information or reoptimization
Multiple
Single
Achievement
Deficit reduction
Pressure uniformity
Time saving
Causal rolling
Causal persistence
Oracle head
-40
-20
0
-60
Paired directional effect vs causal mean (95% scenario bootstrap CI)
c
Scenario-level policy wins
Multiple
Single
Causal rolling 33%
Causal persistence 20%
Causal mean 47%
Oracle head 13%
Causal rolling 50%
Causal persistence 30%
Causal mean 20%
Oracle head 47%
Causal rolling 33%
Causal persistence 23%
Causal mean 43%
Oracle head 13%
Causal rolling 50%
Causal persistence 30%
Causal mean 20%
Oracle head 47%
Causal rolling 27%
Causal persistence 10%
Causal mean 17%
Oracle head 47%
Causal rolling 23%
Causal persistence 27%
Causal mean 17%
Oracle head 33%
Causal rolling 0%
Causal persistence 77%
Causal mean 23%
Oracle head 0%
Causal rolling 0%
Causal persistence 47%
Causal mean 53%
Oracle head 0%
0
50
100
Scenarios in which policy is best (%)